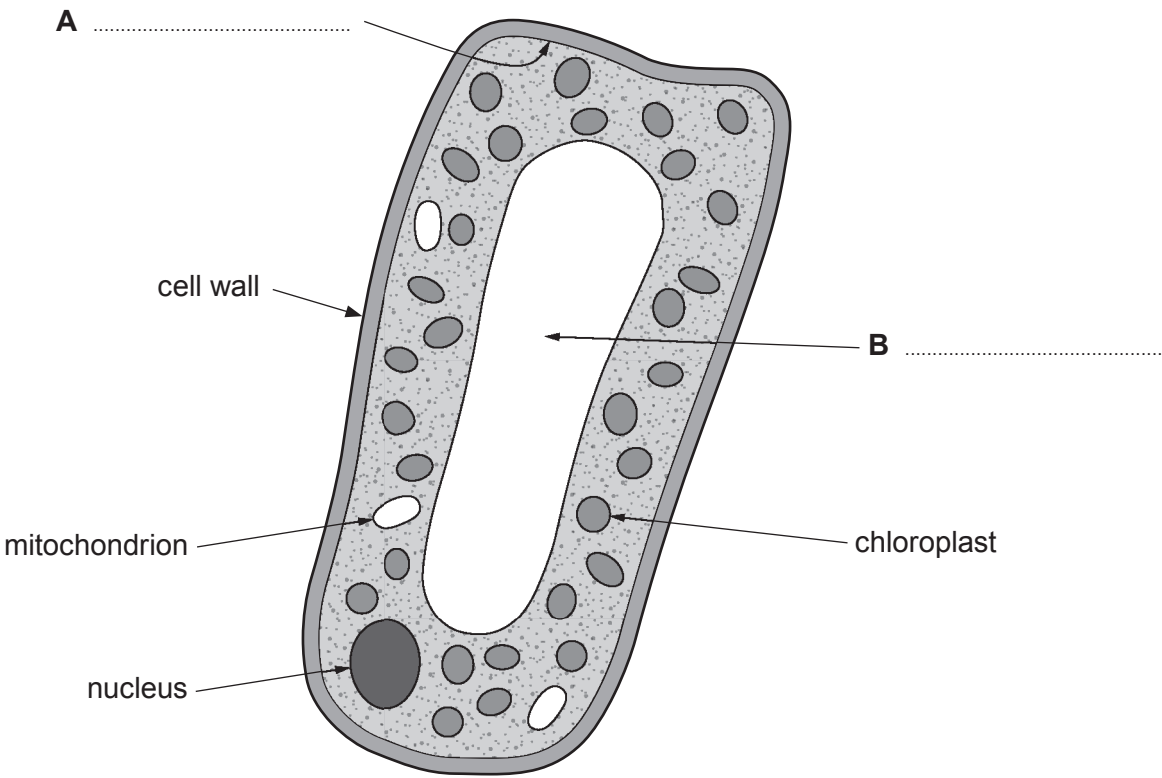


Answer **all** questions.

1. The diagram below shows a section through a plant cell as seen with a light microscope.

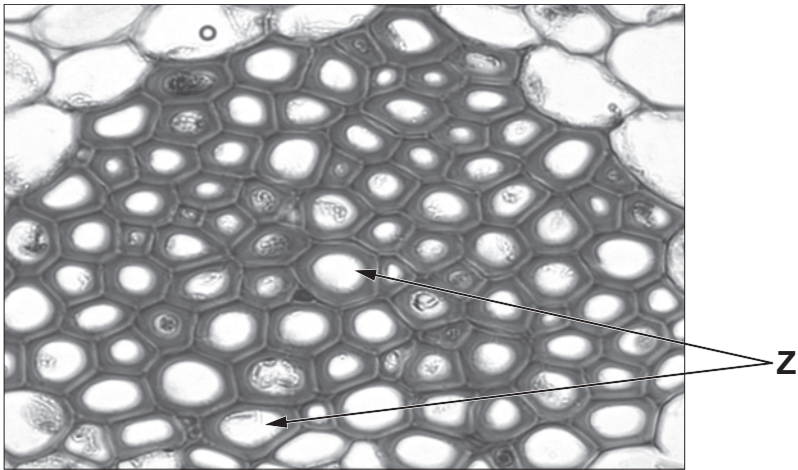


- (a) (i) Label parts **A** and **B** on the diagram. [2]
- (ii) Complete the table below. [3]

Part of cell	Function	Cell part present (✓) or absent (×) in animal cell
.....	photosynthesis	
mitochondrion		✓
.....	contains the chromosomes	

(b) The photograph below shows some plant cells as seen under a light microscope.

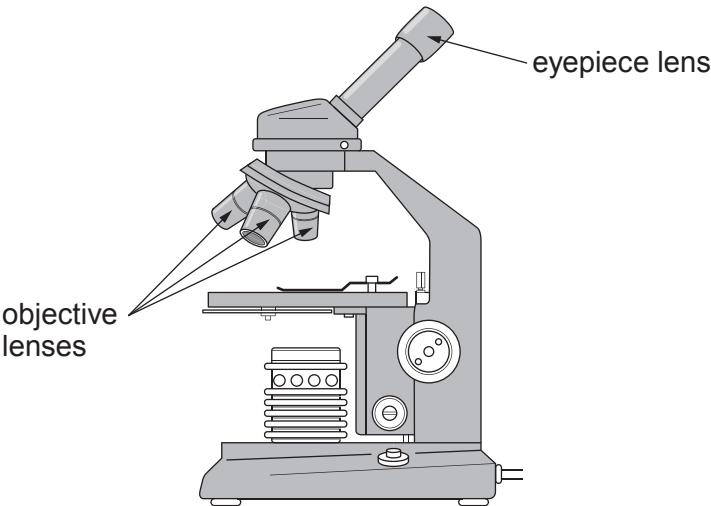
The group of cells labelled **Z** has been treated using a procedure to make their cell walls more clearly visible.



(i) What was the procedure carried out on these plant cells while they were prepared for viewing under the microscope? [1]

.....

(ii) When the microscope was used to view these plant cells the magnification of the eyepiece lens was $\times 10$ and the magnification of the objective lens used was $\times 10$.



Using this information, calculate the total magnification of the image. [1]

Magnification = $\times$

Examiner
only

(iii) State **two** ways in which this investigation could be improved so that the doctors could be more confident that their results were representative of the whole of the population of the UK. [2]

- 1.
- 2.

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- (ii) Name the liquid in layer **A** in the sample and state **two** substances which it transports. [3]

Name of liquid

Substances transported

1.

2.

- (iii) The volume of the blood sample was 20 cm³. Liquid **A** accounts for 45 % of this volume.
Calculate the volume of liquid **A**. [2]

Volume = cm³

- (b) A laboratory technician was working on some blood samples. She noticed that if the samples were left to stand for five minutes before placing them in the centrifuge, they could not be separated properly, as they had started to solidify.

Use your knowledge of the functions of the parts of blood to explain this observation. [2]

.....

.....

.....

(d) State **one** feature of the digestive system which helps this part carry out each function named in (c). [2]

Function 1.

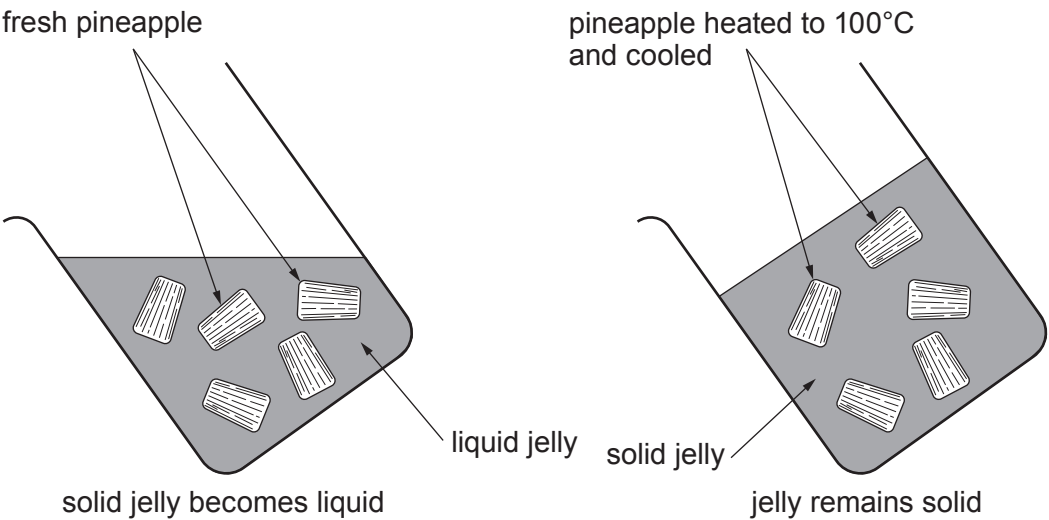
Function 2.

(e) Fresh pineapple is sometimes added to meat to make it tender. Fresh pineapple contains a protease enzyme.

Alan investigated how this took place. He used the following:

- fresh pineapple
- pineapple that had been heated to 100 °C and cooled
- a solid jelly made from protein called gelatin

Two samples of solid jelly were made and the pineapple was pushed into them. After 24 hours the results were shown below:



Explain the results with:

(i) fresh pineapple; [3]

.....

.....

.....

.....

.....

(ii) pineapple heated to 100°C and cooled. [3]

.....

.....

.....

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